

Vasu Bhardwaj

Software Engineer

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Profile

Software Development Engineer with experience in designing secure, enterprise-grade microservices and event-driven architectures. Expertise in orchestrating identity lifecycles (**SCIM 2.0**), implementing **OAuth/OIDC** security flows, and optimizing high-throughput **RESTful APIs**. Strong algorithmic foundation with consistent top-tier performance in competitive programming (**LeetCode Knight**, Rating: 1878)

Professional Experience

Software Engineer

July 2025 – Present
Gurugram

Zenarate

- **Identity & Access Management (SCIM 2.0):** Designed and deployed a SCIM 2.0 identity provisioning microservice integrating Azure AD and Okta, automating user lifecycle management and reducing onboarding time by 60%.
- **Distributed Event Ingestion (Kafka & OAuth):** Built an event-driven ingestion pipeline using **Kafka** and **AWS Lambda** processing 250+ throughput, enabling scalable xAPI analytics ingestion with OAuth-secured service communication.

Education

Kalinga Institute of Industrial Technology

B.Tech in Computer Science & Engineering, 2021 - 25

- CGPA: 8.86 / 10
- Coursework: Operating System, Cloud, Computer Networks, Neural Networks, Object-Oriented Programming

Projects

Incident Commander — LLM Training Environment

Top 100 Finalist · OpenEnv Hackathon (Meta · PyTorch · Hugging Face) Python, FastAPI, Docker, PyTorch, TRL, Unsloth, Hugging Face Spaces | Team VBxAG

- Built a deterministic multi-agent RL environment that trains Llama-3.2-3B to act as an SRE incident commander — triaging alerts, querying observability data, delegating to specialist NPCs, and resolving production outages in a simulated fintech microservices platform under time pressure.
- Designed a three-task difficulty curriculum (deductive canary regression -> third-party attribution -> silent data corruption) with 6-component programmatic grading and structural anti-gaming guards that penalize shortcut strategies by roughly 0.4 score points.
- Ran supervised fine-tuning with 4-bit QLoRA, lifting the model's macro-mean score from 0.693 to 0.815 and more than doubling performance on the hardest task (0.347 -> 0.895).
- Published reproducible training notebook, LoRA adapters, and live deployment on Hugging Face Spaces.

Distributed Task Orchestration Engine

- **Backend Architecture:** Developed a scalable task orchestration engine using **FastAPI** and **Docker**. Implemented **JWT-based stateless authentication** with refresh token rotation to enhance security and session management.
- **Database Design:** Designed a normalised database schema (3NF) for audit logging and task tracking. Optimised query performance using **SQLAlchemy** indexing and connection pooling.
- **Infrastructure:** Orchestrated multi-container architecture using **Docker Compose**, establishing a private network for secure service discovery between the FastAPI backend and MySQL database.

Skills

Languages

Python, C++, JavaScript, SQL

Databases & Infrastructure:

PostgreSQL, MySQL, Kafka, Docker, AWS, Linux, Git

Backend & Security:

FastAPI, Node.js, REST APIs, Microservices, Event-Driven Architecture, OAuth 2.0, OIDC, SCIM

Core CS:

Data Structures & Algorithms, Object-Oriented Design

Certificates

- Microsoft Certified: Azure AI Fundamentals - Jan 2024
- AWS Academy Graduate - Introduction to Cloud